

Algorithms Comparison

An evaluation of Round Robin (Fairness) vs SRTF (Efficiency)

Manual Simulation

Reset Dashboard

Required Test Scenarios

Scenario A

Mixed Workload

Scenario B

Quantum Sensitivity

Scenario C

Short-Job Heavy

Scenario D

Interactive Fairness

Scenario E

Validation Case

Round Robin (RR) Fairness focused

Avg Waiting Time
5.33ms

$\sum WT / n = (5+7+4) / 3$

Avg Turnaround
9.33ms

$\sum TAT / n = (11+13+4) / 3$

Avg Response
2.00ms

$\sum RT / n = (0+2+4) / 3$

Ready Queue Order

P1 P2 P3

Gantt Chart Timeline Q = 3



PID	Arrival	Burst	WT	TAT	RT
P1	0	6	5	11	0
P2	1	4	7	11	2
P3	2	2	4	6	4

Shortest Remaining Time First (SRTF) Efficiency focused

Avg Waiting Time
2.67ms

$\sum WT / n = (0+2+4) / 3$

Avg Turnaround
6.67ms

$\sum TAT / n = (12+6+2) / 3$

Avg Response
0.00ms

$\sum RT / n = (0+0+0) / 3$

Gantt Chart Timeline



PID	Arrival	Burst	WT	TAT	RT
P1	0	6	6	12	0
P2	1	4	2	6	0
P3	2	2	0	2	0

Required Analysis Questions

Which algorithm gave better average waiting time?

SRTF (2.67ms) yielded lower average waiting time than Round Robin (5.33ms).

Did Round Robin appear fairer?

Yes. By using time-slicing (Quantum), it prevents long processes from blocking others indefinitely.

How did the selected quantum affect RR?

Quantum of 3 caused RR turnaround (9.33ms) to exceed SRTF (6.67ms) by 2.67ms. A smaller quantum may improve responsiveness.

Which algorithm gave better response time?

SRTF (0.00ms) yielded lower average response time than Round Robin (2.00ms).

Did SRTF complete short jobs faster?

Yes. Its preemptive nature ensures the process with the least remaining time is always prioritized.

Algorithm Recommendation:

SRTF is recommended — it achieved lower waiting time (2.67ms vs 5.33ms) and lower response time (0.00ms vs 2.00ms) for this workload.

Required Conclusion

- Metric Comparison — WT: SRTF 2.67ms vs RR 5.33ms | TAT: SRTF 6.67ms vs RR 9.33ms | RT: SRTF 0.00ms vs RR 2.00ms. Best overall TAT: SRTF (6.67ms).
- Fairness: Round Robin provided equal time slices to all processes. However, on this workload SRTF matched or outperformed it on response time (0.00ms vs 2.00ms), suggesting the jobs were short enough that SRTF's preemption acted fairly as well.
- Efficiency: SRTF appeared more efficient — it minimised average waiting time (2.67ms vs RR's 5.33ms) by always running the shortest remaining job, reducing total time processes spent idle in the queue.
- Quantum Effect: A quantum of 3 gave RR an average response time of 2.00ms and turnaround of 9.33ms. This quantum balanced fairness and overhead reasonably for this workload.